Experiment 3: Morphological Analysis using NLTK in Python

1. Aim

To write a Python program that demonstrates how to perform morphological analysis using the NLTK library.

2. Procedure

1. Import `PorterStemmer` and `WordNetLemmatizer` from the `nltk.stem` module.
2. Initialize the stemmer and lemmatizer objects.
3. Define a list of inflected words for morphological analysis.
4. Compute the stem and base lemma for each word in the list.
5. Display the original word, its root stem, and its morphological lemma.

3. Program

```
from nltk.stem import PorterStemmer, WordNetLemmatizer

stemmer = PorterStemmer()
lemmatizer = WordNetLemmatizer()

words = ["running", "played", "studies", "leaves"]

for w in words:
    print(w, "-> Stem:", stemmer.stem(w), "| Lemma:", lemmatizer.lemmatize(w, pos='v'))
```

4. Output

```
running -> Stem: run | Lemma: run
played -> Stem: play | Lemma: play
studies -> Stem: studi | Lemma: study
leaves -> Stem: leav | Lemma: leave
```

5. Result

Thus, the Python program was successfully executed to perform morphological analysis using `PorterStemmer` and `WordNetLemmatizer` in NLTK.